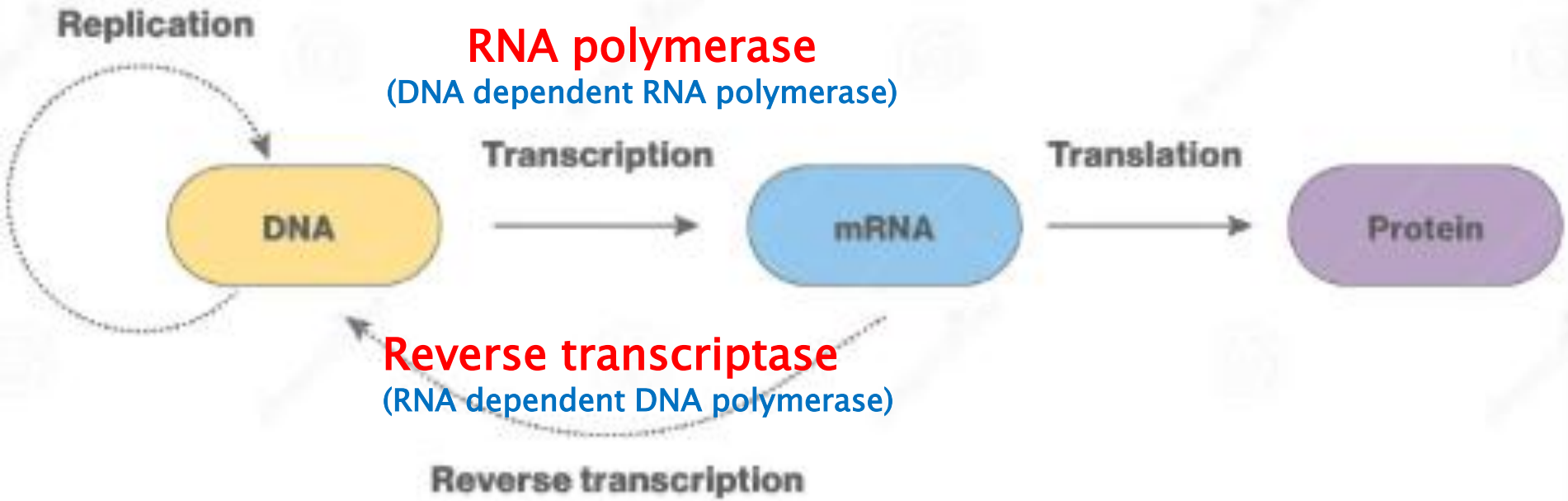
REGULATION OF GENE EXPRESSION

GENE EXPRESSION

- A gene is said to be expressed when the gene product is being produced.
- The flow of genetic information is depicted below:

DNA ⟶ RNA ⟶ PROTEIN ⟶ **Trait**



Regulation of gene expression

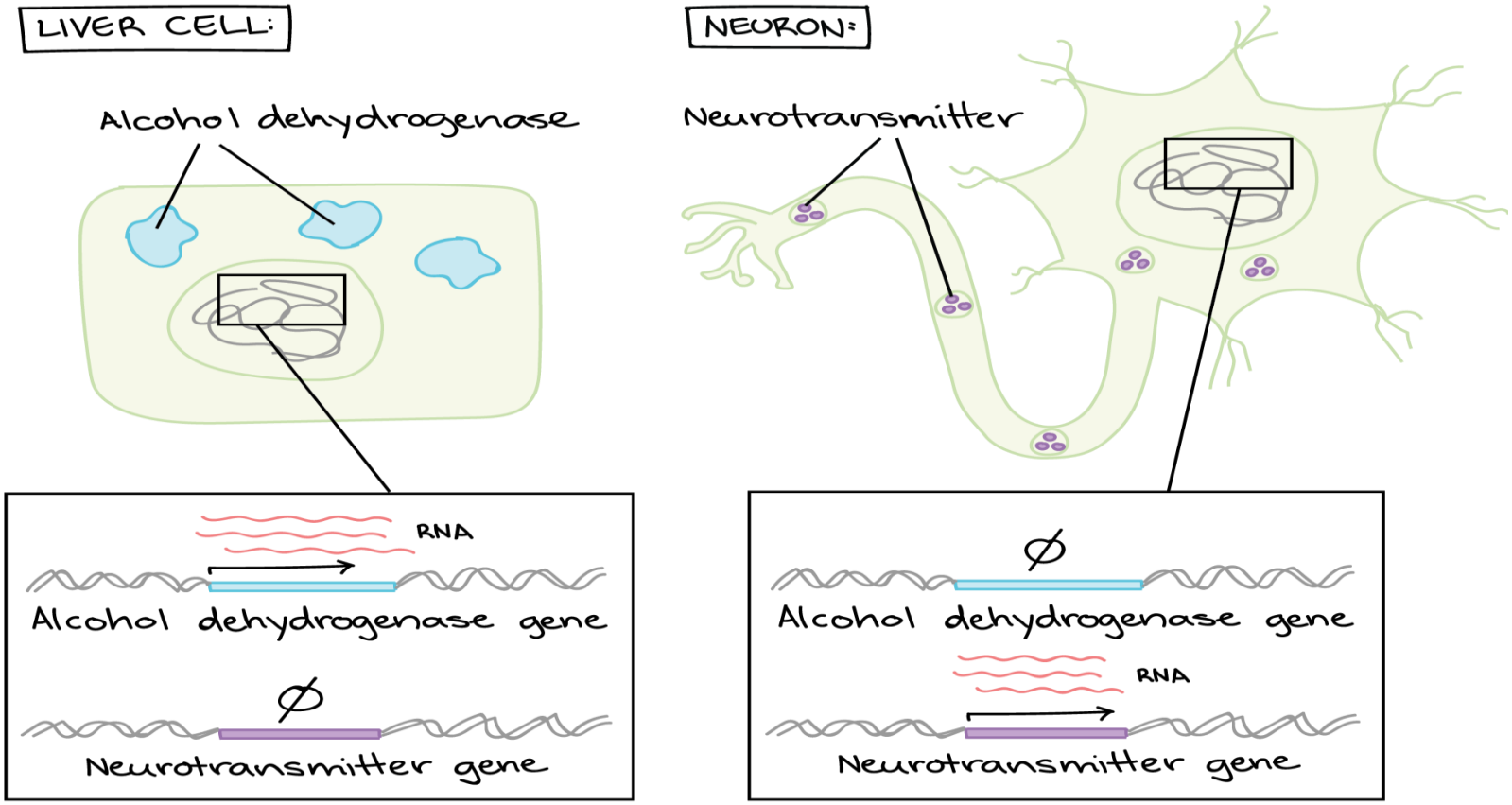
What is gene regulation and why it is required?

- A single human cell contains over 100000 different genes. Yet only a fraction of those genes are expressed in any one cell at a given time.
- **Gene regulation refers to** the ability of cells to determine which genes should be expressed, when those genes should be expressed and the extent to which each gene should be expressed.
- Potential control points where gene expression can be turned on or off, speed up or slowed down
- The cells of all multicellular organisms are specialized to perform the specific functions of the tissues in which they reside.

Differential gene expression

At any given time, a typical human cell expresses about 20% of its genes.

Enzyme alcohol dehydrogenase is expressed in liver cell, but expression of neurotransmitter gene is switched off. However, in neurons, dehydrogenase enzyme expression is switched off, but neurotransmitter genes are expressed.



Classification of gene with respect to their Expression:

- **Constitutive (house keeping) genes:**
- 1- Are expressed at a fixed rate, irrespective to the cell condition.
- 2- Their structure is simpler
- ***Controllable genes:*** **Or Non-constitutive genes**
- 1- Are expressed only as needed. Their amount may increase or decrease with respect to their basal level in different condition.
- 2- Their structure is relatively complicated with some response elements

TYPES OF REGULATION OF GENE

POSITIVE REGULATION :

- When the expression of genetic information is quantitatively increased by the presence of specific regulatory element is known as positive regulation.
- Element modulating positive regulation is known as activator or positive regulator.

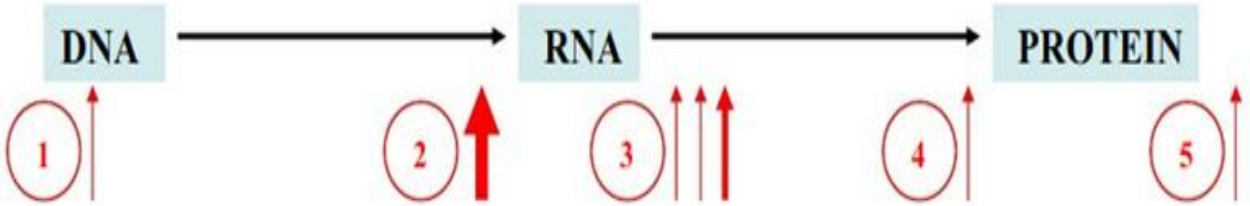
NEGATIVE REGUATION:

- When the expression of genetic information is diminished by the presence of specific regulatory element is known as negative regulation.
- The element or molecule mediating the negative regulation is said to be repressor.

The regulation of gene expression is necessary for the growth, development, differentiation and existence of organism.

Gene regulation

CONTROL LEVELS OF GENE EXPRESSION IN EUKARYOTES



1. *Control at the level of chromatin and genome structure* → **Gene regulation at chromatin level**
2. *Control at the level of transcription initiation* → **Transcriptional regulation**
3. *Control at the level of post-transcription initiation including transcription elongation, mRNA stability, alternative splicing* → **Post-transcriptional regulation**
4. *Translational control* → **Translational regulation**
5. *Post-translational control* → **Post-translational regulation**

Five levels of gene regulation

Potential control points where gene expression can be turned on or off, speed up or slowed down

1) **Chromatin remodeling:** the region of DNA must be opened for enzymes and transcription factors to access the gene. DNA is wrapped in nucleosomes (histone proteins + DNA complexes) known as chromatin. Histone proteins undergo modifications for opening of DNA for transcription of genes.

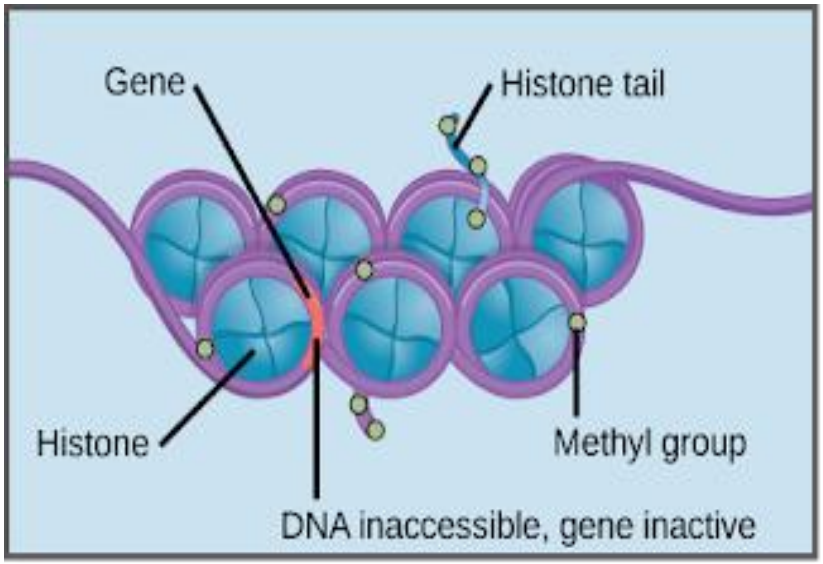
2) **Transcriptional regulation:** Determination of whether or not, and to what extent, a gene will be transcribed.

3) **Post-transcriptional regulation:** Regulation of the processing, transport, and longevity of mRNAs. In eukaryotes this level includes regulation of the rate and extent of processing pre-mRNA into mature mRNA, the transport of mRNA into the cytoplasm and the rate of degradation.

4) **Translational regulation:** Determination of whether mRNA will be translated or not and extent of translation.

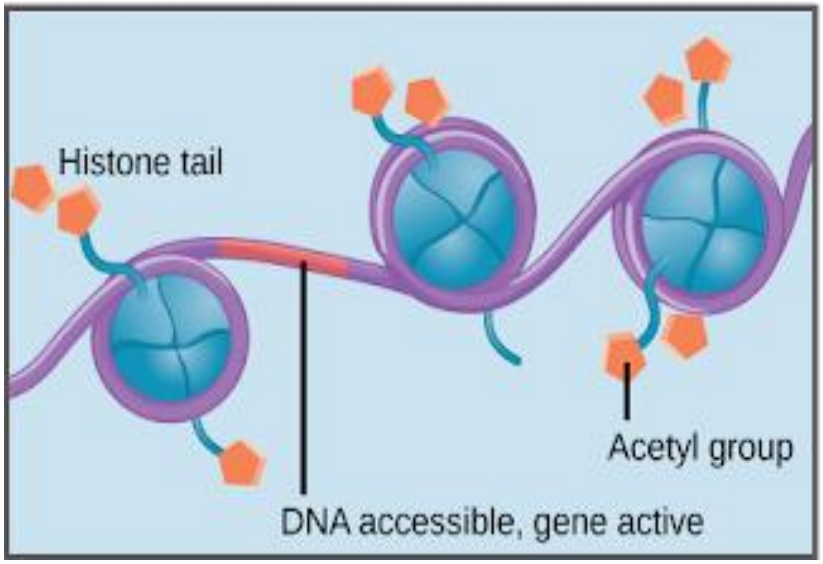
5) **Post-translational regulation**: Regulation of protein modification and activity (includes modification of inactive protein to form active protein)

1. Gene regulation at chromatin level



Methylation of DNA and histones causes nucleosomes to pack tightly together. Transcription factors cannot bind the DNA, and genes are not expressed.

Methylation of histones – genes are not expressed as genes are inaccessible by transcription factors.



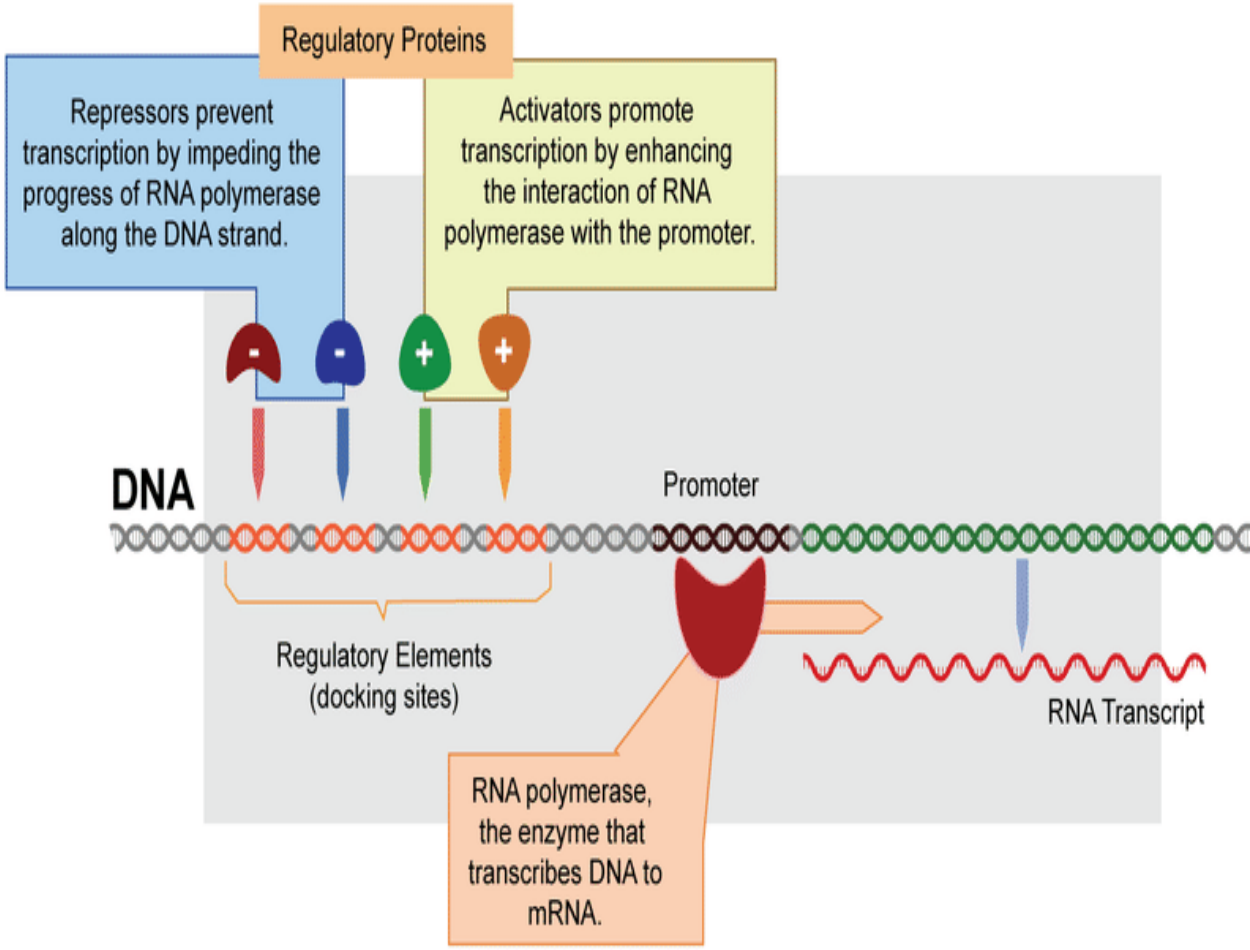
Histone acetylation results in loose packing of nucleosomes. Transcription factors can bind the DNA and genes are expressed.

Acetylation of histones – genes are expressed by easy accessibility of genes by transcription factors.

2. Transcriptional regulation: Determination of whether or not, and to what extent , a gene will be transcribed

Repressors: are transcription factors that has a **DNA-binding domain**, which interacts with specific DNA sequences, and a **repressor domain,** which interacts with other proteins (regulatory proteins to repress the transcription).

Activators: are transcription factors that has a **DNA-binding domain**, which interacts with specific DNA sequences, and an **activation domain**, which interacts with other proteins to stimulate transcription from a nearby promoter.



Transcriptional gene regulation and its components

***Regulatory proteins- repressors and activators in transcriptional regulation

3. Post-transcriptional gene regulation

❖ Regulation of the processing, transport, and longevity of mRNAs.

In eukaryotes, this level includes regulation of the

1. Rate and **extent of processing of pre-mRNA into mature mRNA**,

❖ **Pre-mRNA processing to mRNA: 5'-capping; polyadenylation, splicing (removal of introns)**

2. Alternative splicing in nucleus

3. Rate of degradation of mRNA

❖**Stability of mRNA in the cytoplasm**

Post-transcriptional gene regulation

- ❖ **Alternative splicing in nucleus**
- ❖ **Stability of mRNA in the cytoplasm**

- Alternative splicing: is a very common phenomenon in higher eukaryotes. It is a way to get more than one protein product out of the same gene and a way to control gene expression in cells.

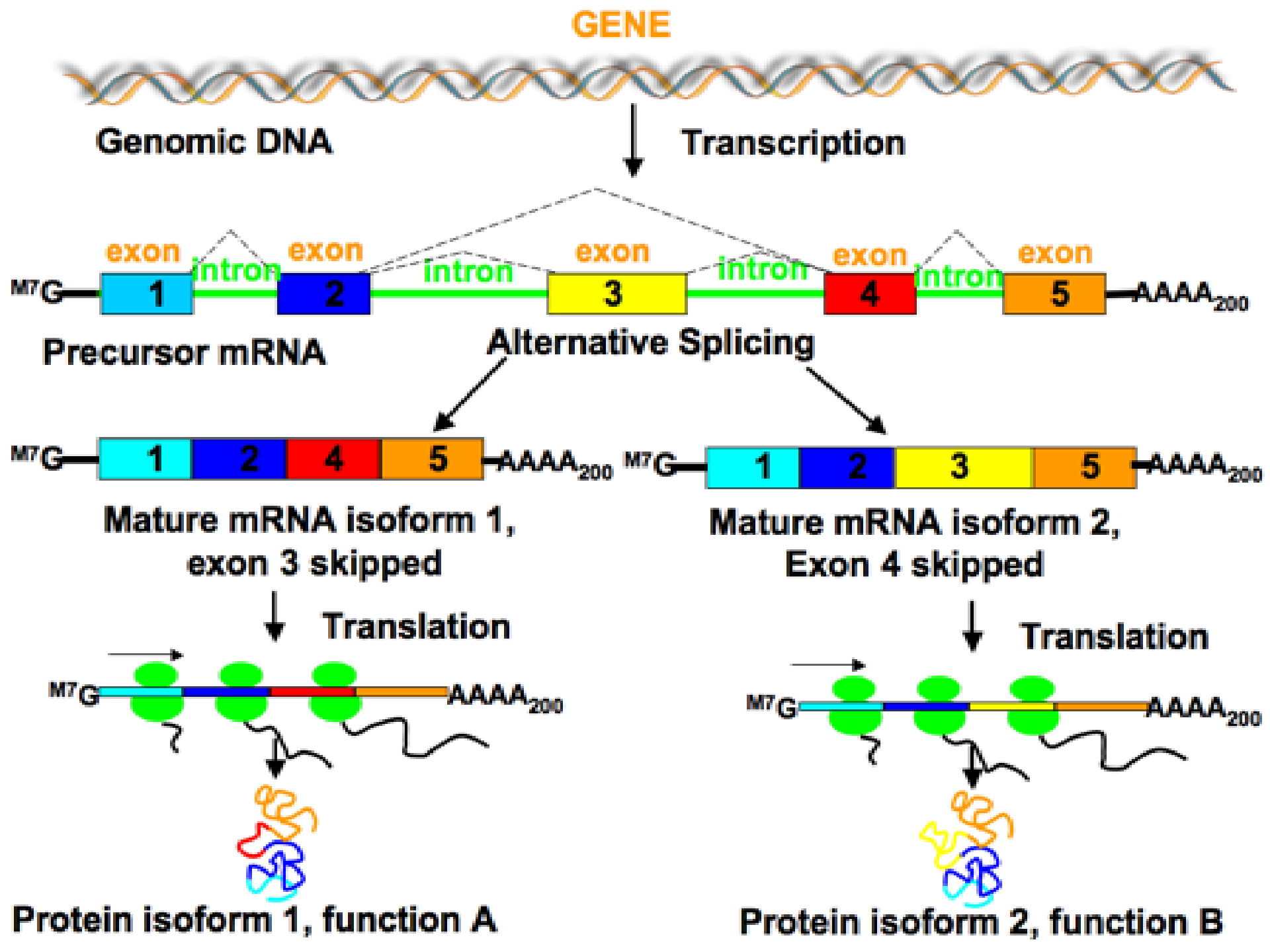


Exons joining

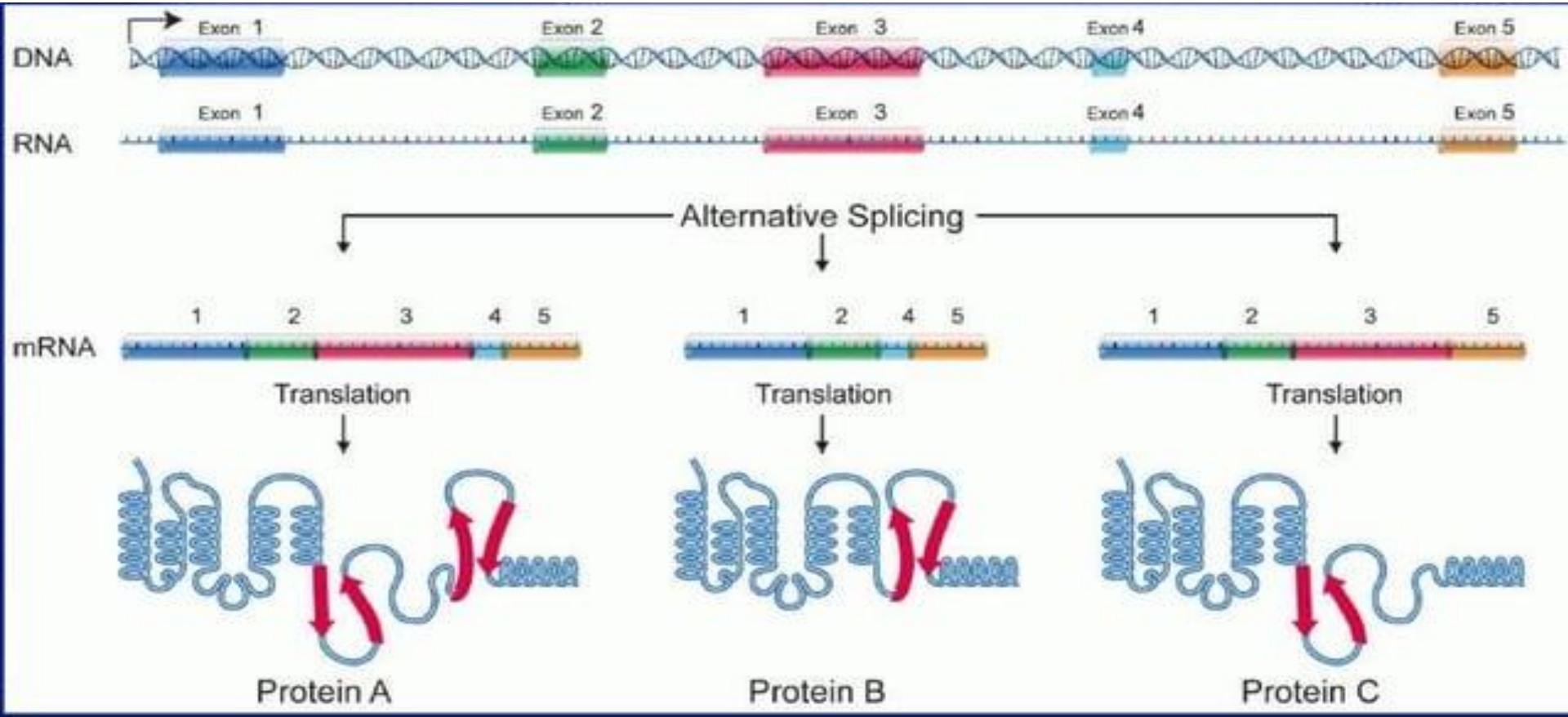
- Joining of 500+100+300+800 = **1700 bases**
- Joining of 500+200+800 = **1500 bases**
- Joining of 500+200+300+800 =**1800 bases**

Intron's removal

- Removal of 200 base (exon) with intron
- Removal of 100 and 300 base exon with intron
- Removal of 100 base exon with intron


GENE
Genomic DNA
Transcription
exon
intron
exon
intron
exon
intron
exon
intron
exon
M7G
1
2
3
4
5
AAAA200
Precursor mRNA
Alternative Splicing
M7G
1
2
4
5
AAAA200
Mature mRNA isoform 1,
exon 3 skipped
M7G
1
2
3
5
AAAA200
Mature mRNA isoform 2,
Exon 4 skipped
Translation
M7G
AAAA200
Translation
M7G
AAAA200
Protein isoform 1, function A
Protein isoform 2, function B

Alternative splicing (or differential splicing) is a process by which the exons of the RNA produced by transcription of a gene a primary gene transcript or pre-mRNA are reconnected in multiple ways during RNA splicing.



- 60% of the genes in the human genome are spliced in alternative ways to generate more than one protein per gene.

Stability of specific mRNA in the cytoplasm

- Some mRNAs are unstable and have a very short life span in the cytoplasm, whereas others may be much more stable, permitting them to be translated many times.
- **The production of a gene product may be regulated by altering the stability of a specific mRNA.**
- **For example, the production of casein, an abundant protein in milk.**
- **The hormone Prolactin, increases the stability of mRNAs that encode casein**, causing increased translation of casein when prolactin is present.

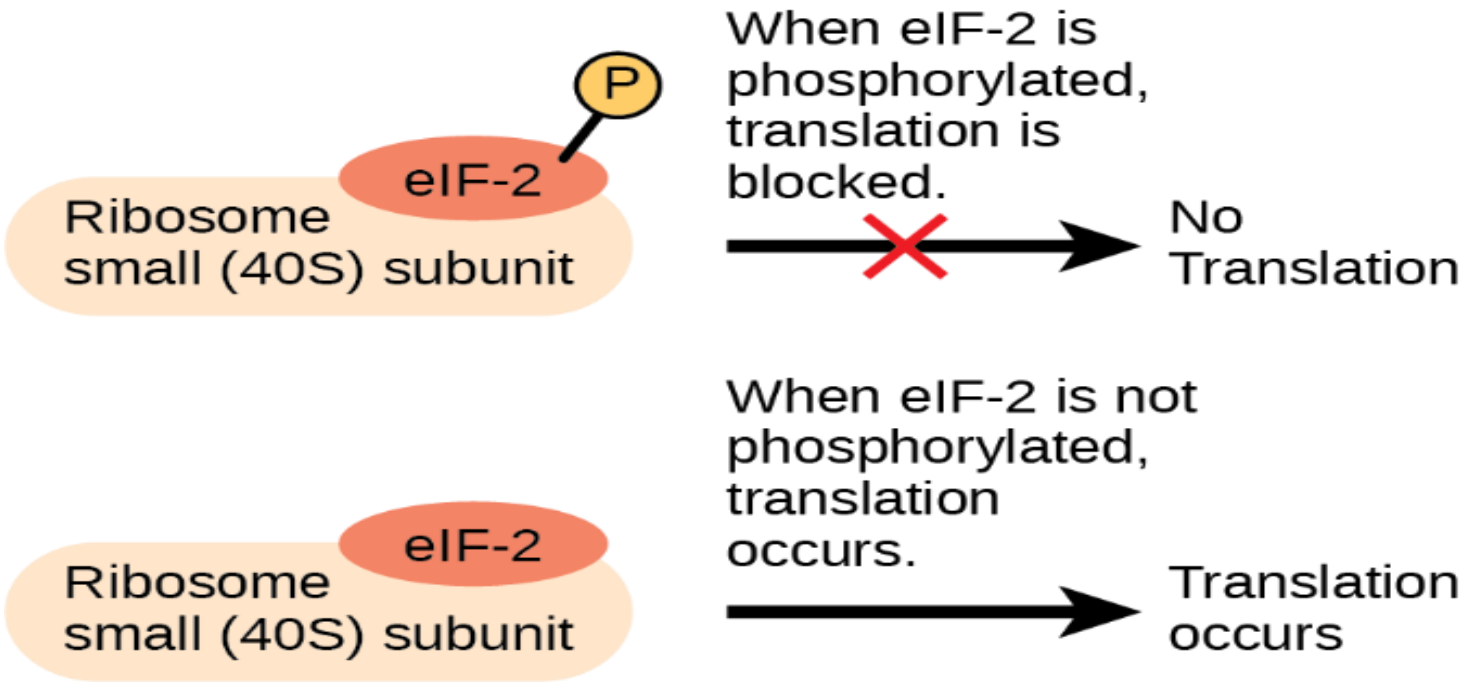
4. Regulation at the level of translation

- In translational control, regulators select which mRNAs in the cytoplasm will go for translation by ribosomes

Mechanisms of translational control:

- **Transport control:** only the mRNA which is transported to the cytoplasm can be translated.
- **Number of ribosomes**
- **Translational factors:** If the concentration of these factors is too low in the cell, the translation initation, or the elongation process can be inhibited.

Translational control involved initiation factors that bind the translation initiation complex.



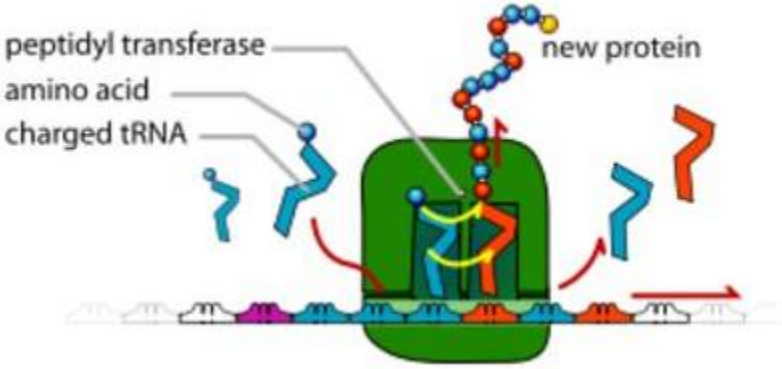
An increase in phosphorylation levels of eIF-2 has been observed in patients with neurodegenerative diseases such as Alzheimer's, Parkinson's, and Huntington's

5. Post translational regulation

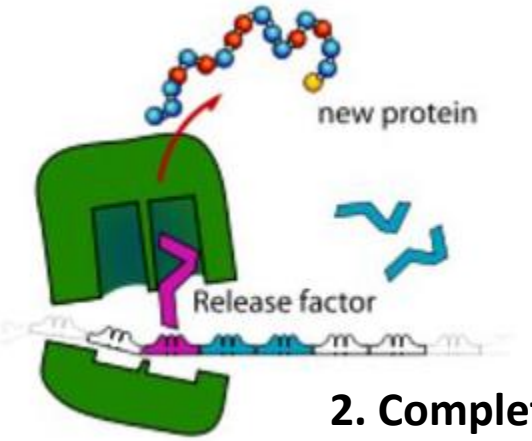
- **Post translational regulation refers** to the control of levels of active protein. It is performed by means of reversible events such as posttranslational modifications and irreversible events such as proteolysis.
- **Chemical modifications of proteins** after its translation is called post translational modifications. **The chemical modifications that takes place** at certain amino acid residues after the protein is synthesized.
- Mostly occurs in endoplasmic reticulum and golgi apparatus.

Importance of post translational modifications include:

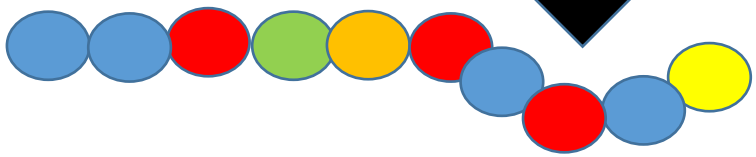
- Stability of proteins
- Generating heterogeneity in proteins
- Help in utilizing identical proteins for different cellular functions in different cell types
- Translocation of proteins across biological membranes to their target location for activity, in cell signalling interactions
- Contribute towards biological processes and diseased conditions.



1. Translation process-decoding the codon message into polypeptide formation

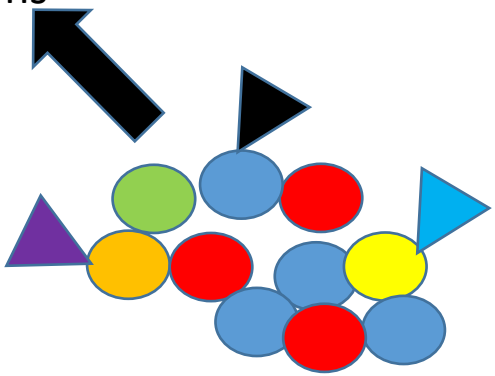


2. Completion of Translation process



3. Inactive newly synthesized protein released from translational machinery

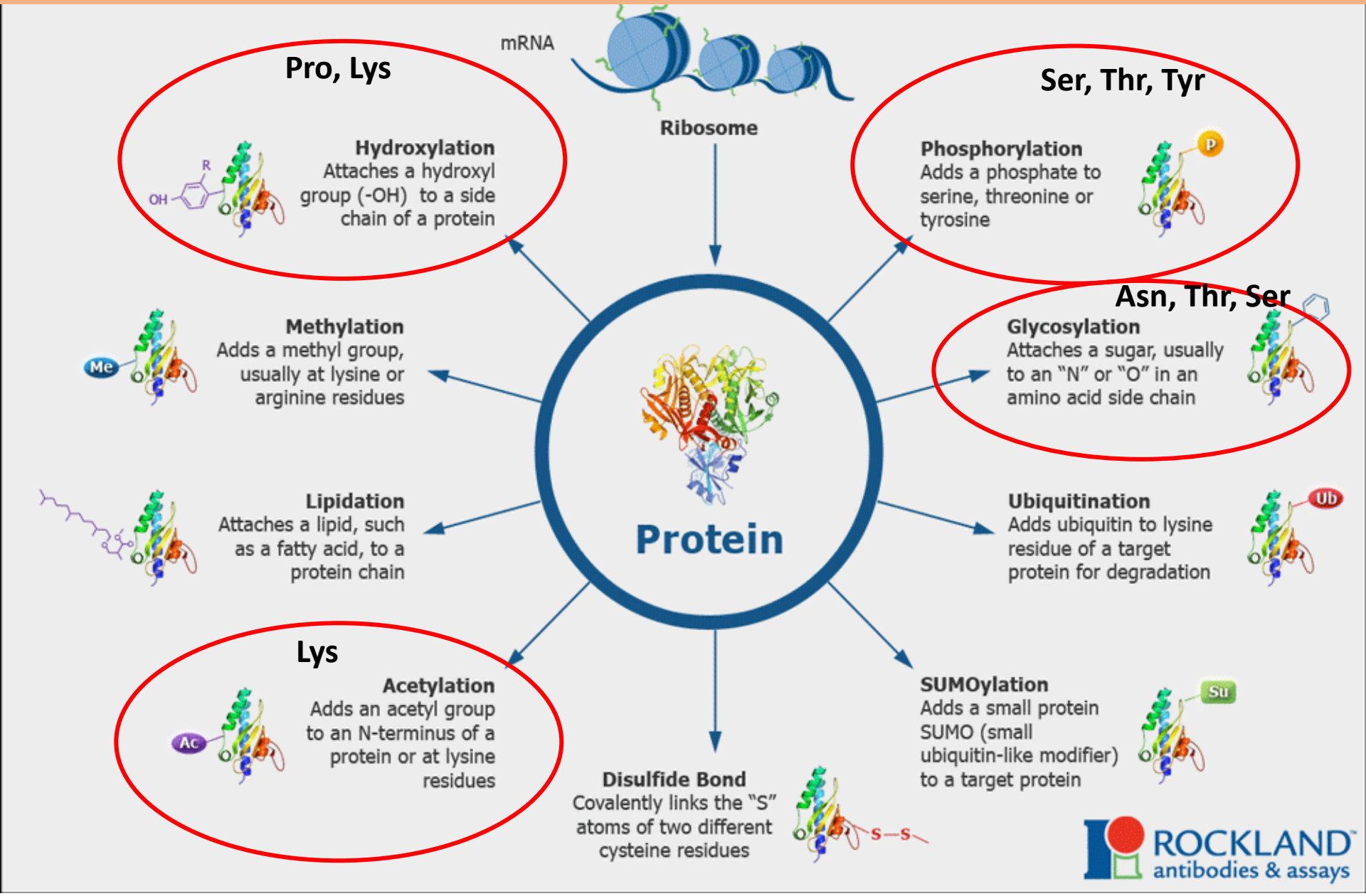
4. After post translational modifications, attachment of chemical groups, conformational changes and folding



5. Active and functional protein after post translational modifications

6. Active protein targeted to Specific location for carrying out biological functions

Attachment of chemical groups as post translational modifications



Post translational regulation as a modification involving peptide bonds cleavage (limited proteolysis)

Peptide Bonds Cleavage (limited proteolysis, specific and well-regulated)

- Signal leader peptide removed by signal peptidase (both in prokaryotes and eukaryotes)

  Precursor protein → mature protein (Insulin)
- Zymogen → active enzyme
- Trypsinogen → Trypsin
- Pepsinogen → Pepsin
- Prohormone → Hormone
- Polyprotein → neuropeptides (peptide hormone) conversion